

Peripheral neural interfaces for reading high-frequency brain signals

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Jaime Ibáñez^{1,2,3}✉, Blanka Zicher⁴, Etienne Burdet⁴, Stuart N. Baker⁵, Carsten Mehring^{6,7} & Dario Farina⁴✉

Accurate and robust recording and decoding from the central nervous system (CNS) is essential for advances in human–machine interfacing. Technologies for direct measurements of CNS activity are limited by their resolution, sensitivity to interference and invasiveness. Motor neurons (MNs) represent the motor output layer of the CNS, receiving and sampling signals from different regions in the nervous system and generating the neural commands that control muscles. Muscle recordings and deep learning decode the spiking activity of spinal MNs in real time and with high accuracy. The input signals to MNs can be estimated from MN outputs. Here we argue that peripheral neural interfaces using muscle sensors represent a promising, non-invasive approach to estimate some of the neural activity from the CNS that reaches the MNs but does not directly modulate force production. We discuss the evidence supporting this concept and the advances needed to consolidate and test MN-based CNS interfaces in controlled and real-world settings.

Neural signals from the central nervous system (CNS) reach the muscles and control when and how they generate forces to produce movement. The motor unit is the key structure in the translation of neural information into muscle contraction. A motor unit consists of a spinal motor neuron (MN) and the group of muscle fibres that it innervates and controls¹. Motor units represent the frontier between neural processing and behaviour, and MNs are the last common neural pathway controlling behaviour². The activation of each MN in the spinal cord is mirrored by the activation of the innervated muscle unit, thus the muscle unit can serve as an accessible proxy of neural activity. From an engineering perspective, reading the activity of MNs provides access to the output of the somatic motor nervous system (Fig. 1a) and a way to infer its working principles.

Currently, methods for directly—that is, invasively—recording representative ongoing activity from the CNS are scarce. However, there is growing evidence of increasingly safe approaches, such as intracranial recordings used in various medical devices, which have

led to meaningful advancements in fields such as speech prostheses and other types of brain–machine interface for patients with severe paralysis^{3,4}. Nonetheless, exploring alternative methods for recording CNS activity remains crucial to address existing limitations in current technologies and to further enhance safety and efficacy in neurotechnologies⁵. As a potential alternative to direct brain recordings, we can use an inverse modelling approach to investigate signals in the CNS by decoding the final electrical activity produced at its outputs, in the periphery of the nervous system.

Advances in sensors and signal processing methods, including deep learning methods, have enabled access to the neural activity of sets of MNs by recording the activity they evoke in muscles using electromyography (EMG)⁶. EMG recordings represent a mixture of the electrical activity from the muscle units that compose a muscle, which serve as proxies for the MNs in the spinal cord. EMG signals are therefore recordings that embed the final output of the CNS. Identifying motor unit activity in the EMG signals then yields a neural interface.

¹BSiCoS Group, Departamento de Ingeniería Electrónica y Comunicaciones, Instituto de Investigación en Ingeniería de Aragón (I3A), Universidad de Zaragoza, Zaragoza, Spain. ²BSiCoS Group, Instituto de Investigación Sanitaria (IIS) Aragón, Zaragoza, Spain. ³Centro de Investigación Biomédica en Red de Bioingeniería, Biomateriales y Nanomedicina (CIBER-BBN), Zaragoza, Spain. ⁴Department of Bioengineering, Imperial College, London, UK. ⁵Institute of Neuroscience, Newcastle University, Newcastle upon Tyne, UK. ⁶Bernstein Center Freiburg, University of Freiburg, Freiburg im Breisgau, Germany. ⁷BrainLinks-BrainTools, University of Freiburg, Freiburg im Breisgau, Germany. ✉e-mail: jibanez@unizar.es; d.farina@imperial.ac.uk

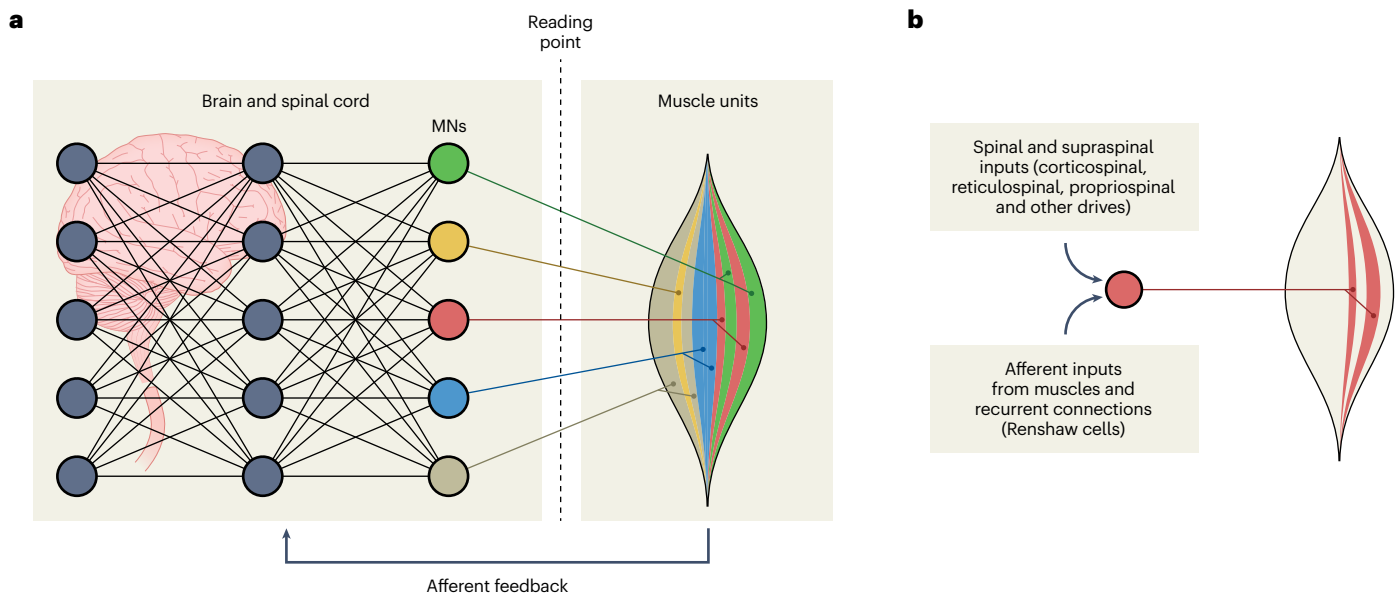


Fig. 1 | Convergent inputs to MNs. a, MNs in the muscles represent the output layer of the CNS. By measuring the electrical activity that muscles produce during contraction, we can characterize the firing patterns of MNs and therefore read the neural activity of this last layer of the nervous system complex network.

b, MNs receive inputs from multiple sites in the CNS (including direct brain projections) and from sensory organs in the periphery. All this information drives MN activity⁷.

This concept is schematically described in Fig. 1a. In this scheme, the CNS and peripheral nervous system constitute a large neural network with high complexity. This network projects its activity to MNs, which form the last layer of the network and connect with different parts of the CNS⁷ (Fig. 1b).

By characterizing the MN activity that reaches muscles, we can access some of the neural signals most tightly associated with behaviour. This has been a key reason for the use of myocontrol for interfaces in clinical applications, such as using EMG to estimate motion intent and to control active prosthetics⁸. However, it is unclear whether the motor command to the muscles (that is, the neural inputs to muscles that determine or modulate the level of contraction) is the only information that MNs encode, or whether the MNs also encode activity in the CNS that does not directly modulate muscle force. This question relates to how much information about neural processing within the CNS can be extracted from MN recordings.

As MNs control muscles, one would think that their output is tightly matched to muscle characteristics so that the full information transfer from MNs to muscles is associated with force production. However, numerous studies have shown that MN discharge patterns, and therefore EMG signals, contain information that is related to brain oscillations but not directly to force production. For example, it is well known that cortical and EMG signals are correlated in the beta band (13–30 Hz)^{9–11}, which lies outside the muscle contraction dynamics^{12,13} and thus information in this band cannot directly influence force, at least within a linear approximation of force production.

These classical observations raise questions about which cortical oscillations are transmitted to the output of the CNS (the MNs) and why. They also suggest that part of the neural activity generated in the brain, containing relevant information regarding brain states and dynamics, may be observed in the electric fields produced by muscles, even if it is not associated with underlying muscle contractile activity. This is because there is a mismatch between the bandwidth of the electrical and mechanical activity of muscles: muscle electrical signals have a broader spectral bandwidth than mechanical muscle contraction; thus, they potentially contain more information. EMG may offer a window through which we can look at brain activity.

This Perspective discusses existing evidence supporting the possibility of using the activity of MN populations to estimate neural activity originating in the CNS that is not directly involved in force production. This introduces the concept of brain interfacing via muscle sensors, which differs from classic EMG-based interfaces aimed at decoding the motor command sent to the muscles to generate an intended action or gesture. Along these lines, we envision a technology that senses the electric fields in muscles and processes them to infer informative brain signals, such as circuit-specific cortical oscillations at different frequencies in patients with a brain lesion, or changes in identified brain signals that can be used to develop models and strategies for human–machine interaction.

The following sections introduce the key concepts that allow the decomposition of large populations of MNs using non-invasive or minimally invasive sensors, and the physiological basis supporting the notion that this neural information can be used to infer ongoing neural activity in relevant regions of the CNS, such as the motor cortex. Finally, we propose frameworks in which MN-derived information can be used to advance the fields of neuroimaging and neural interfacing and improve our understanding of the neural control of movement.

Muscle interfaces to decode the output of the CNS

The contribution of a motor unit to the electric potential recorded at a certain point on a muscle can be modelled as a train of delta functions convolved with the motor-unit-specific action potential waveform, as measured from that recording point¹⁴. The EMG recording thus corresponds to the sum of the electric potentials produced by all the active motor units and noise components.

In this model, the trains of delta functions associated with each motor unit are the neural activity (that is, the output of the CNS) that can be estimated through the deconvolution of the EMG signal (Fig. 2). The possibility of identifying this neural activity from EMG signals depends on two key properties of the biological and biophysical systems underlying muscle activity. The first property is the stability of the synaptic connection between each MN and its muscle unit. This connection means that each action potential travelling along the axon of an MN determines a corresponding action potential in the innervated muscle fibres, that is, in the corresponding muscle unit¹⁵.

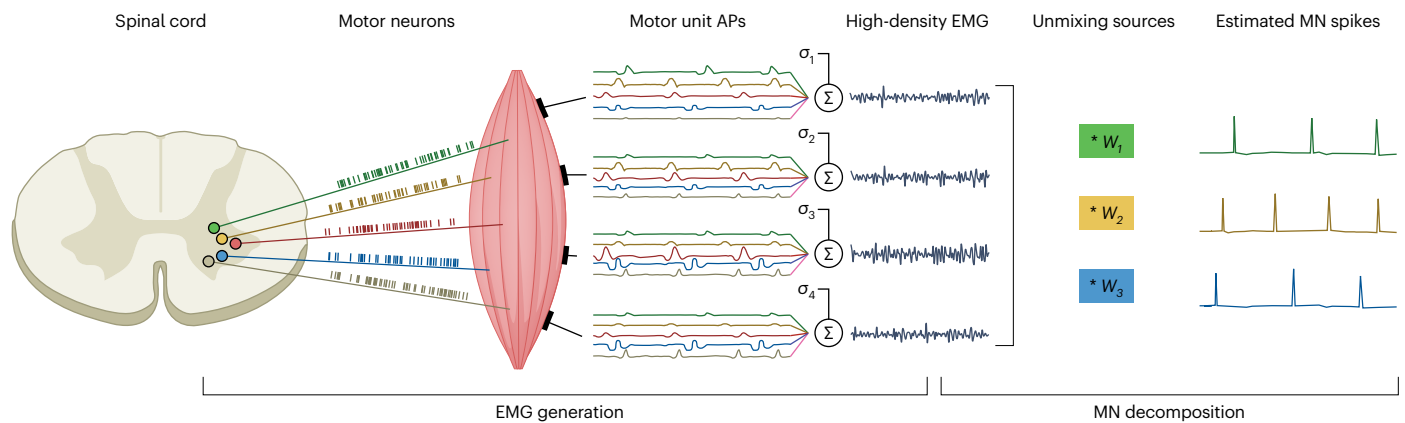


Fig. 2 | Simplified model of EMG signal generation and the decomposition of MN activity from multichannel EMG with classic deconvolution methods.

The cell body of MNs is in the ventral horn of the spinal cord. On the basis of the synaptic inputs MNs receive, they generate trains of action potentials (APs) that travel along their axons to the muscles. Each MN AP causes a corresponding muscle fibre AP that travels along the muscle fibres of the motor unit formed with the innervating MN. Placing EMG electrodes at different points on the muscle

results in recordings of the muscle fibre APs generated by different units, each with distinct waveforms (based on the relative position of the electrode and the recorded fibres), along with additional independent noise of standard deviation σ . By multiplying the EMG recordings by unmixing matrices (the ' W_i ' terms), we can enhance the time points at which specific motor units generate APs travelling along the muscle fibres. This provides a reliable read-out of the spiking activity of MNs corresponding to the identified motor units.

This one-to-one relation between MNs and muscle unit action potentials implies that the spiking activity of single neurons can be measured by identifying the action potentials produced by muscle fibres of a motor unit. The second property is the spatial and temporal differentiation of the electrical activity that muscle fibres produce in the surrounding volume conductor, that is, the motor unit electrical signatures. Each motor unit is unique within the muscle in terms of the spatial distribution of muscle fibres, neuromuscular junctions and fibre membrane characteristics. This implies that the convolutive filters are unique for each motor unit, at least under ideal conditions of sampling in space and time.

Within a linear and stationary model, the two properties mentioned previously and the sparsity of spiking in individual MNs allow us to solve the deconvolution problem and to isolate the activity of individual MNs^{6,16}.

The deconvolution of neural activity from invasive or non-invasive EMG recordings can be solved by blind source separation methods customized to the convolutive nature of the mixtures and the sparsity of the sources (Fig. 2). Alternatively, deep neural networks¹⁷ or generative models of EMG signals and their inverse formulation^{18,19} can be used. The current main aim in the development of more refined methods for EMG deconvolution is to increase the number of decoded MNs, which can be attained both by refinement in electrode technology and by advances in decoding algorithms²⁰.

The performance of current conventional methods to decode EMG signals strongly depends on the number of neural sources relative to the number of observations and on the stability of the action potential waveforms of individual motor units. Because of the large number of motor units in a muscle, only some of them can be accurately identified by current source separation methods. Decoding is typically limited to the sources with the greatest energy and with distinctive motor unit action potential shapes (if two motor units produce action potentials with comparable temporal-spatial shapes, as measured from the recording points, they will not be distinguishable)²¹. In addition, the ability to decompose sources is limited in all contractions that involve changes in joint angles, which applies to most natural tasks. In fact, changes in muscle length and relative position with respect to the skin during motion modify the characteristics of the volume conductor separating the sources from the recording electrodes^{22,23}.

Artificial-intelligence-based approaches may be used to mitigate the above limitations. An improvement over conventional statistical

source separation methods can be obtained by training deep networks to perform the decomposition process, using deconvolution methods for labelling and then increasing the robustness of the network-based decomposition by extensive data augmentation. Current models of EMG signal generation are sufficiently realistic to provide simulated data useful for augmenting neural-network-based decomposition of experimental data. For example, recent studies have shown that a realistic model of EMG signal generation can augment the training of a network based on gate recurrent units that learn how to decompose EMG signals, making the resultant decomposition more robust than the original deconvolution algorithm used for labelling²⁴. Moreover, a hybrid architecture that combines elements of deep latent variable models and conditional adversarial training can provide a generative model of EMG signals that can mimic the outputs of complex numerical models of human muscle biophysics. Such a framework can be used to continuously sample from a dynamically changing system in real time¹⁸. This allows us to model all factors of variability in the generation of EMG signals and use them for data augmentation, and to derive inverse models that estimate internal parameters (such as MN discharge patterns) based on observations.

The decomposition process can be performed in a fully unsupervised way by identifying the embedded neural code in a reduced dimensional space, imposing constraints on the properties of this space. For example, a proposed autoencoder learns the discharge times of individual motor units by processing multichannel EMG recordings²⁵. Following preprocessing of the EMG recording via convolutive sphering, the encoder is constrained to apply an orthogonal transformation to the spatial data and the optimization criterion enforces a temporal sparsity constraint. Because of its encoding-decoding structure, this method does not need to be trained on known decomposition results.

MNs linearly transmit high-frequency neural signals

Spinal MNs receive a barrage of synaptic inputs from afferent nerves coding proprioceptive and other sensory information, and from spinal and supraspinal circuits, including direct inputs from the cerebral cortex through monosynaptic corticospinal projections²⁶. All these inputs are integrated and translated into spiking activity that is transmitted to muscles (Fig. 1b). Therefore, the twitches that muscle fibres produce can be seen as the result of a complex transformation of the net excitatory inputs that MNs receive from different sources in the

CNS. The CNS ensures that the appropriate neural inputs are received by MNs to generate an intended movement.

The net excitatory input received by MNs is typically modelled as a set of independent and common synaptic inputs²⁷. Independent inputs represent the synaptic inputs to MNs that are not correlated with the inputs received by other MNs in a pool. Conversely, common synaptic inputs are those that are the same as or correlated to inputs that (many) other MNs in the pool receive^{28–30}. Owing to the differences in the biophysical properties of MNs within a pool, their responses to inputs tend to linearize and amplify the common inputs relative to the independent inputs when the neural activity of MNs is analysed at the population level^{31–35}. For this reason, a group of MNs receiving and sampling common and independent inputs can be viewed as a neural linear amplifier of the common inputs, with very high rejection of independent noise³⁶. This is evident when considering the layer of MNs as a single-layer perceptron that acts as a linear regressor if the activation functions are linear. In the biological case, this corresponds to considering the MNs as active approximately in the middle of their input–output response curve³⁷. We can therefore conclude that the last layer of the CNS is an approximately linear one that amplifies the common inputs it receives when considering the activity of populations of MNs that are fully recruited and not reaching saturation³⁸. The amount of amplification of the linear components by a group of MNs will increase as a function of the group size³⁶. Indeed, the observation that common inputs to MNs act as a linearization process aligns with population coding principles that are also observed elsewhere in the nervous system^{39,40}. In the retina for instance, decoding the activity of small populations of retinal ganglion cells rapidly saturates information extraction, indicating that a few neurons can effectively represent the encoded information when it is commonly distributed⁴¹. Similar results have been observed in the auditory system using multiple spike trains in auditory receptor neurons⁴².

The linearization performed by MNs applies to any common inputs they receive; groups of MNs can effectively transmit multiple sources of information at different frequencies simultaneously³⁵. Figure 3 exemplifies this property of MNs in a simulation in which a group of MNs receives common inputs at different frequencies along with independent inputs (different for each MN). Simple band-pass filtering of the sum of the spike trains of all MNs allows a reliable estimation and separation of the common inputs at multiple frequencies.

The conversion from motor unit action potentials into muscle force can be modelled as a second-order critically damped filter, which strongly attenuates high-frequency neural inputs to muscles (here we consider high-frequency inputs to muscles as those above ~10 Hz)^{12,13,43,44}. Indeed, the frequency of movements and forces produced by limbs are constrained to rather low frequencies, with (pathological) tremors representing the highest frequencies that can be measured in the arms and hands^{45,46}. However, and somewhat surprisingly, the neural inputs that MNs receive, and therefore their outputs, go far beyond the maximum working frequencies of muscles. This can be experimentally observed when decoding a large number of MNs that can show common activity at frequencies >70–80 Hz (ref. 20). This neural information may consist of different components, spanning various frequency ranges, originating from diverse regions of the nervous system that project into the MN pools. Given the characteristics of how muscle transforms neural inputs into forces, this broad spectrum of common inputs suggests that a substantial portion of the common synaptic inputs to MNs will be transmitted along their axons without having a direct influence on muscle forces (Fig. 4). These high-frequency inputs may be associated with ongoing activity in the CNS, reaching and modulating the firing patterns of MNs. Because of the linear transmission by MN populations, extraction of this high-frequency activity, which does not directly affect forces, can be achieved using standard linear separation methods (Fig. 3).

Accessing brain activity through MNs

Identifying the sources contributing to the high-frequency activity in MNs and understanding why this activity reaches MNs are relevant challenges to advance in the framework of MN-based CNS interfaces. Given the limitations of non-invasive neuroimaging technologies to record signals originating in the CNS, the interpretation of the high-frequency activity in the MNs is complex, particularly due to the lack of reference signals guiding the analysis. By simultaneously recording brain and muscle activity, many studies have consistently shown that the oscillatory activities at 20–30 Hz (beta rhythms) and 40–50 Hz (low gamma rhythms) carried by MNs have a cortical origin^{9,10,47–54}. This implies that muscles are to some extent driven by the cortical rhythms that are encoded by MNs. For MNs to provide valuable information about the ongoing cortical activity that they receive, the characteristics of the corticospinal transmission of such signals must be stable over time. Also, to implement neural interfaces with the brain with MNs, transmission needs to be established with relatively small delays, and modulations of cortical oscillatory inputs to MNs need to be strong enough to be clearly sensed from MNs.

Neurons in the corticospinal tract, the main descending pathway connecting the brain and MNs, are entrained by ongoing activity in neighbouring areas of the motor cortex, such as beta and gamma rhythms^{55,56}. Although highly heterogeneous and complex pathways connect cortical neurons and MNs, cortical neural activity is effectively and reliably transmitted to the muscles through the fastest and most straightforward pathways because of the distribution of transmission delays across the thousands of channels connecting the motor cortex to MNs⁵⁷. This implies that MNs can provide a timely and stable read-out of ongoing cortical neural activity. Figure 5a shows an example of the linear association, measured using spectral coherence⁵⁸, between cortical and MN frequency components, specifically beta rhythms that are prominent during sustained contractions. On average, there is a time delay of approximately 28 ms between cortical signals and MN activity (Fig. 5b), which corresponds to the expected transmission delays in the fastest pathways connecting the motor cortex and the measured muscle, in this example, the tibialis anterior muscle. These results show the fast transmission of cortical rhythmic activity to the muscles, supporting the idea that decoding MN from muscle sensors enables interfacing with the motor cortex.

The use of muscle recordings to characterize the functional connection between the brain and MNs, as well as to estimate brain signals indirectly, can be further examined using neurofeedback paradigms. For example, neurofeedback paradigms led participants to modify the coherence amplitudes in the beta band during sustained contractions, without changing the elicited forces⁵⁹. Other studies used real-time decoding of EMG signals into MN activities to train participants to volitionally modulate rhythmic beta activity (13–30 Hz) in the neural drive to muscles^{60,61}. After a relatively short training process, this muscle-based neurofeedback paradigm led to changes in the beta activity detected in muscles and in cortical electroencephalography (EEG). Results further indicated that beta events (or bursts) identified in the muscles during sustained contractions had a similar duration and rate of detection of bursts to beta events identified from the motor cortex, suggesting that the beta activity transmitted by MNs has a primarily supraspinal origin. Investigating optimal ways to deliver feedback to participants may allow us to gain a deeper insight into the characteristics of the high-frequency neural signals sampled by MNs, study how well participants can learn to modulate high-frequency inputs to muscles during different types of motor tasks, and generate models explaining the functional roles of these types of input to muscles.

When recordings of muscle activity occur during long periods of steady muscle contractions, the observed fluctuations in cortical rhythmic activity projected to MNs are moderate but strong enough to be detected in relatively small groups of MNs; for example, around 12 motor units were identified and tracked in real time per participant in

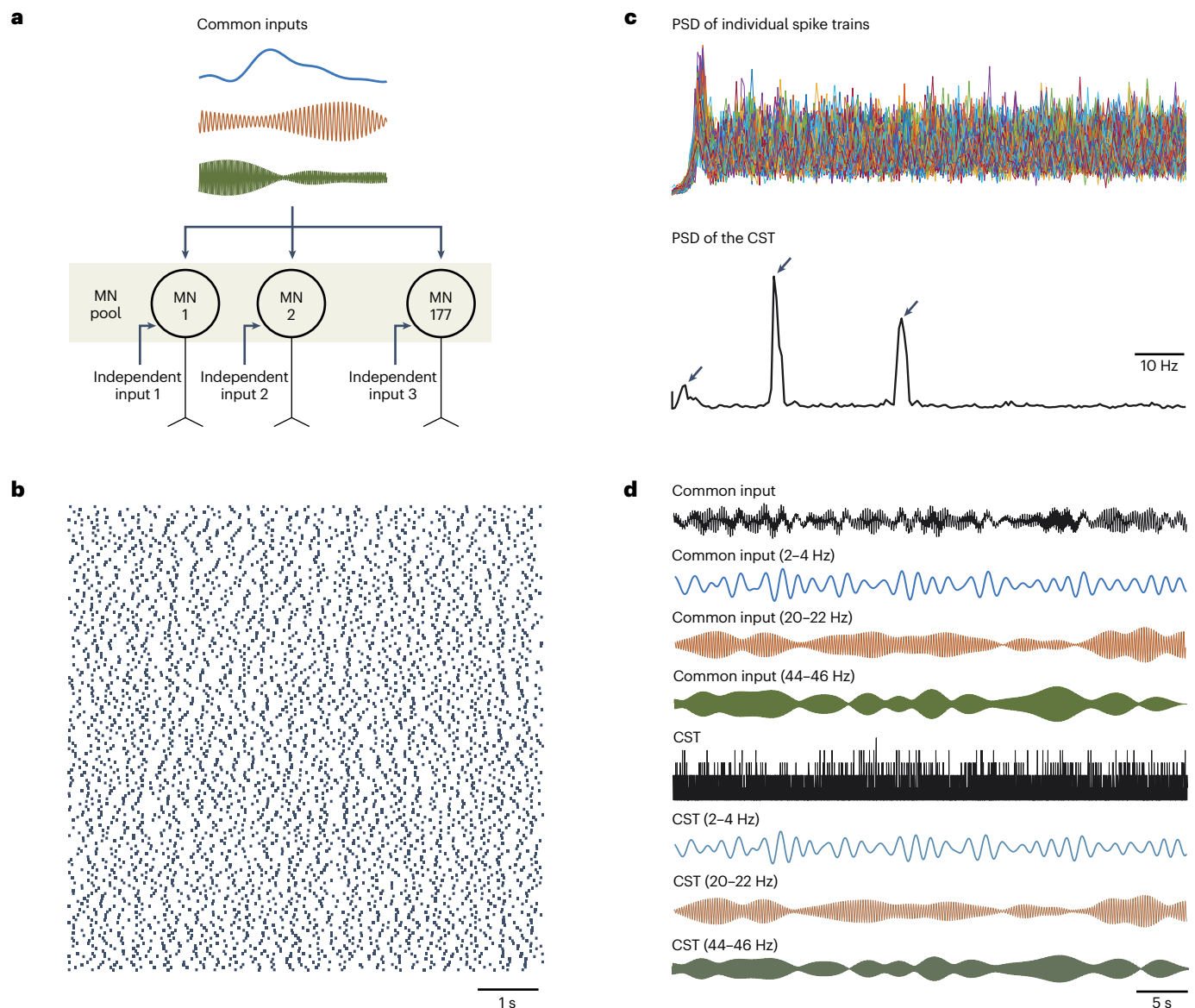


Fig. 3 | Linearization of common input transmission by MNs. **a**, Simulation of a group of 177 MNs receiving a common input (sum of 3 inputs at 2–4 Hz, 21 Hz and 45 Hz) plus an independent input that is different for each MN. **b**, Raster plot with the spike trains of the simulated MNs. Each row represents the spiking activity of a simulated MN. **c**, Power spectral densities (PSDs) of the spike trains produced by the individual MNs (top) and of the composite spike train (sum of all spike trains; bottom). The mean of the signals was subtracted to obtain the PSD. The arrows represent the peaks in the PSD at the frequencies of the common inputs. **d**, The common input to the MNs and the band-pass-filtered versions of the common input at different frequencies (top four plots) and the resulting cumulative spike train (CST; bottom four plots) and its filtered version at the frequencies of the components contributing to the common inputs. To simulate the results

in this figure, a previously validated model was used^{27,91}. The model simulated the spiking activity of 177 MNs driven by common and independent inputs. The common input was a signal obtained by summing white Gaussian noise filtered in 3 bands (2–4 Hz, 20–22 Hz and 44–46 Hz). The three filtered signals were normalized in power before summing them. The common input also included a direct current component so that the simulated number of inputs arriving from the common projection per 0.2-ms time step was 1.7 (which is equivalent to each MN receiving 8,500 impulses per second). In addition to the common input, MNs also received independent inputs that were modelled as white Gaussian noise with mean equal to the variance. The level of the independent input was adjusted to make MNs fire at physiologically realistic levels during low-force contractions.

the neurofeedback experiments³¹. To further investigate how strongly the power of high-frequency (>10 Hz) cortical projections to MNs can fluctuate and how these changes can alter corticospinal transmission, it becomes relevant to use experimental approaches that induce strong changes in high-frequency motor cortical oscillations during steady contractions^{62–64}. A study⁶⁵ assessed the magnitude and consistency of changes in the spectral contents of motor unit population activity during a movement preparation and cancellation paradigm. Two types of trial were designed for this, ‘go’ and ‘no-go’ trials, guided by auditory cues. During the trials, participants holding a tonic contraction

of ankle dorsiflexion were first given a warning stimulus to prepare a motor response (ballistic contraction), and 1 s later they received an imperative stimulus that required them to either release the prepared action (go trials) or to cancel the preparation but keep holding the stable contraction (no-go trials). Cancellation of a prepared action is known to be associated with strong increases in cortical beta activity, which may have a role in preventing changes in motor cortical outputs^{64,66}. The study showed that while participants successfully held the sustained contractions in the no-go trials, salient changes in power across different spectral bands in the MNs could be observed

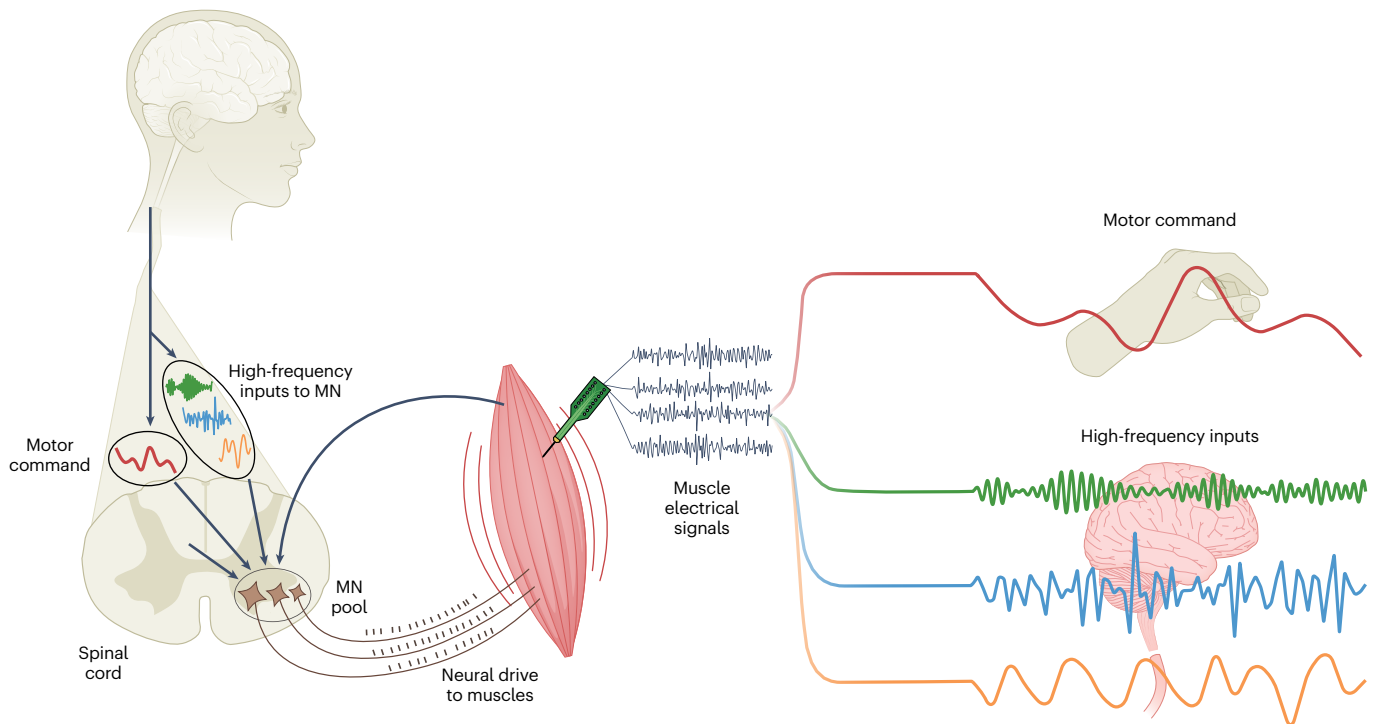


Fig. 4 | Separation of common synaptic inputs to MNs. MN pools can sample multiple common synaptic inputs. The low-frequency inputs control the level of muscle contraction to determine movement^{12,43,44}. The high-frequency inputs do not interfere with movement directly but they are still encoded by the trains of action potentials travelling along MNs and muscle fibres and can be decoded.

(see data from a representative participant in Fig. 6a). Some of these changes (between 13–30 Hz and 30–45 Hz) were related to changes in cortical activity occurring in parallel with the cancellation of the prepared action in the no-go trials⁶⁵. These results provide evidence of successful corticospinal transmission of salient cortical oscillations without changes in motor outputs. This further supports the concept that MNs can simultaneously control force production and receive, and be driven by, high-frequency inputs from distal areas in the CNS. Further support for this idea can be found in a study⁶⁷ that provides consistent evidence of corticomuscular transmission of beta bursts occurring without any detectable changes in the force produced by tonically contracted muscles (Fig. 6b).

The previous studies suggest that MN activity may allow us to distinguish brain states related to different types of mental processing. In another study⁶⁸, MN activity was decoded during steady contractions while participants performed different types of mental task (right foot motor imagery, imagining movements of both hands, mathematical mental operation or not focusing on any specific tasks). Using a filter bank and a linear classifier, the mental tasks could be classified in some participants with moderate to low accuracies, using only high-frequency components in the MN activity as input features for classification. Furthermore, the modulations in high-frequency components were incompatible with changes in force level. This study replicated classic brain-interfacing approaches based on imagery with peripheral recordings, showing that some mental task-induced oscillations from supraspinal areas may leak down to MNs, allowing the discrimination of mental tasks above chance level via EMG. This study also involved imagining non-motor tasks, further supporting the notion that MN activity may be used to retrieve information about ongoing brain activity, even if it is not associated with motor processing.

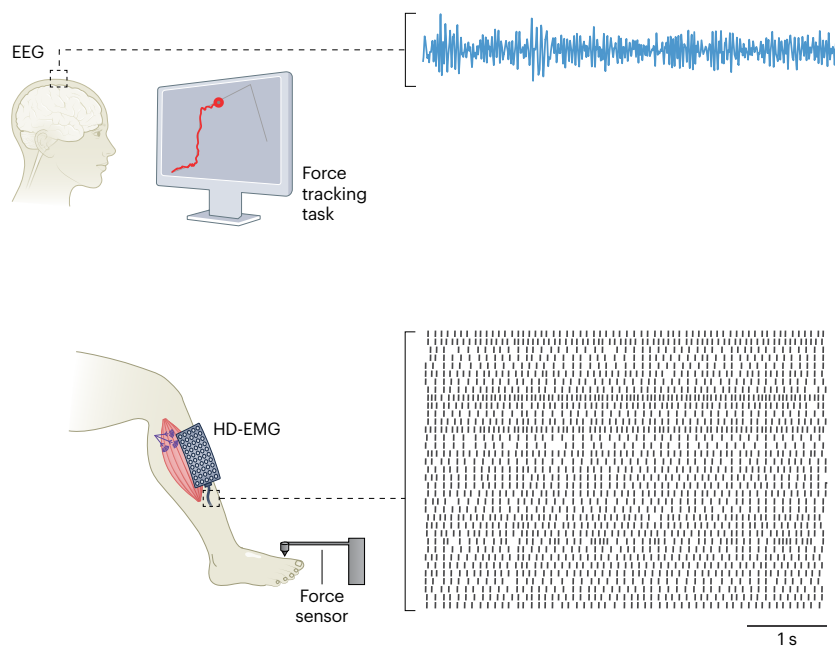
Overall, there is consistent evidence that MNs receive and encode rhythmic activity that does not modulate forces and that have origins in the brain, for example, in the motor cortex. To further advance in the interpretation and use of high-frequency neural inputs to MNs,

it becomes critical that multiple common inputs to MNs can be separated. This should be possible using standard spectral separation techniques if the different neural inputs to MNs are independent and have non-overlapping spectral bandwidths. However, this may not be expected if neural inputs have large bandwidths or if they are only transmitted to MNs through relatively small neural populations. In such a case, the inputs will present harmonic components that can be summed with neural sources in other frequency bands, with nonlinear mechanisms. An alternative possibility in this context is to use other source separation methods that rely on the statistical properties of the inputs to be decoded. These methods may take advantage of the fact that inputs to MNs arising from different parts of the CNS may have distinct patterns of projections to MNs within a muscle or to the groups of muscles they innervate^{69,70}. Alternatively, spike prediction models may be implemented to estimate upstream neural activity from downstream spiking behaviour^{71–73}. A recently proposed reinforcement-learning-based point process framework has been used to generatively predict spike trains through behaviour-level rewards⁷⁴, that is, by using behavioural information to reduce the complexity of estimating neural activity in higher centres from downstream activity. The framework could be used to estimate the neural information associated with movement from the spiking activity of MNs encoding cortical information and, as a consequence, the activity that does not influence movement.

Prospects of the peripheral neural interfaces with the CNS

The ability to decode CNS signals from MN activity presents an exciting opportunity to access neural information that would otherwise be difficult to obtain non-invasively in unconstrained contexts. This is especially important for understanding neural activity in deep brain structures and spinal circuits, which are not currently measurable with existing technologies but may project their activity onto MNs in specific motor contexts.

a Brain and muscle recordings during isometric contractions



b Directed coherence

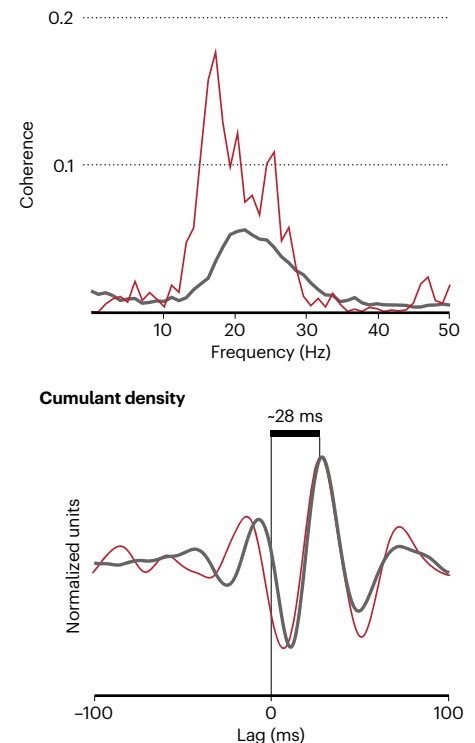


Fig. 5 | Stable transmission of cortical activity to muscles. Simultaneous recordings of brain and muscle activity to assess their neural connectivity. **a**, The EEG signal from the primary motor cortex (blue trace) and high-density EMG (HD-EMG) recording from the tibialis anterior muscle were acquired while the ankle produced an isometric flexion force (at approximately 10% of the maximum force that the study participant could generate). The MN activity decoded from the HD-EMG recording was analysed together with the EEG recordings from the scalp area above the cortical representation of the studied muscle. **b**, Corticomuscular coherence in the descending direction (brain to

muscle; top) provides evidence for the linear transmission of cortical activity at ~20 Hz to MNs. The analysis of the temporal association between the cortical and MN activity (bottom) indicates a lag between the correlated activity in the 2 recordings of -28 ms, which corresponds to the average brain–muscle transmission delays through the fastest neural routes connecting the brain and the tibialis anterior⁹². The red traces indicate the data from one representative participant; the grey traces are the average group data ($n = 19$ participants). Panel **b** adapted from ref. ⁵⁷ under a Creative Commons license [CC BY 4.0](https://creativecommons.org/licenses/by/4.0/).

The possibility of using MNs as an indirect measure of the human brain is promising in the development of neural interfaces. Having access to large populations of MNs during a variety of motor conditions is expected to lead to improved estimates of high-frequency signals encoded by MNs and to a more refined understanding of how these signals co-vary with behaviour. Neural interfaces based on MN recordings may provide access to both spinal and brain signals, as long as the connectivity between circuits in the CNS and the muscles is not disrupted by a lesion. With modern EMG recording and processing techniques, we can now track the activity of large volumes of MNs with increasing precision and generalizability, both across participants and out of the laboratory, making this a viable approach for man–machine interfaces^{20,22,75,76}. This type of interfacing can be accomplished with minimally invasive or even non-invasive devices, allowing for potentially large information transfer by simultaneously and accurately measuring multiple neurons. It also requires minimal calibration and may achieve long-term stability in tracking individual motor units across multiple sessions if the position of the EMG electrodes is maintained relative to the measured muscles^{77,78}.

Using MNs to develop an interface with the human brain may be particularly interesting for expanding human motor abilities by allowing people to control artificial limbs concurrently with their natural limbs⁷⁹. The difficulty in realizing such motor augmentation is to identify neural signals that can be controlled independently from and in coordination with natural motor control signals⁸⁰, and to transform the extracted information that is outside the motor space into a

command that can allow a meaningful control of an electronic device⁸¹. MNs can be measured using wearable sensors and may transmit simultaneously low- and high-frequency neural contents to muscles. If some of the high-frequency signals transmitted to muscles are not directly linked to motor function, they may represent a resource for expanding the number of degrees of freedom that humans can control without interfering with the movements of natural limbs, which could allow movement of different parts in the body (see concept in Fig. 7).

As a step towards this goal, real-time decomposition of MN activity allowed participants to control two components of the spectral content encoded by MNs⁶⁰. After brief periods of training, participants could partially dissociate the activity in the two frequency bands that they had to control independently. This provides proof of concept that it is possible to extend the degrees of freedom that a single muscle can control (from one to two in this case). To advance this line of research, work is now needed to identify methods to extract and classify high-frequency signals in MNs and develop optimal training protocols that enable humans to control neural processes originating in the brain and transmitting through the muscles. One promising direction would involve the data-driven extraction of latent variables from MN behaviour that are orthogonal to movement kinematics. This could be done using artificial-intelligence-based approaches applied to dynamic modelling and latent factor analysis of MN populations^{82–84}. Approaches that enable the identification of behaviourally relevant subspaces from cortical activity could be adapted to dynamic systems at the spinal level using MN data decomposed from muscle recordings.

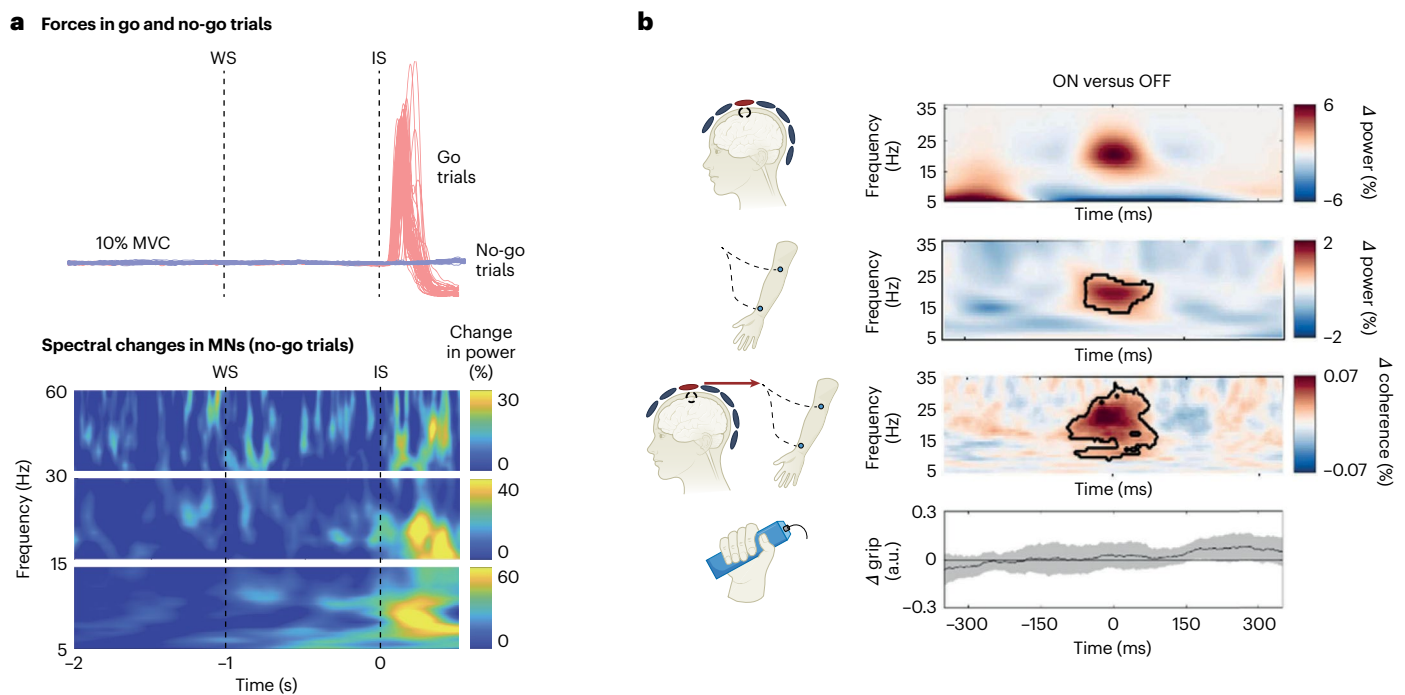


Fig. 6 | Evidence for changes in high-frequency signals sampled by MNs that do not translate into force changes during isometric contractions. **a**, Changes in different frequency bands in the MN activity during the preparation and cancellation of a motor task in a go/no-go paradigm. Top: force traces from a representative participant in a study using a go/no-go paradigm to measure changes in high-frequency MN activity. In each trial, participants had to mildly activate the tibialis anterior muscle (producing a dorsiflexion force of 10% of the maximum voluntary contraction (MVC) achievable by each participant). After some time keeping the force stable, participants were presented with two cues: a warning stimulus (WS) informed about the upcoming imperative stimulus (IS), which was given 1 s after the WS. The IS could either mean that participants had to perform a fast dorsiflexion with the minimum reaction time possible (go trials) or that they had to remain in the same position without changing the produced force with the ankle (no-go trials). Bottom: results from the same representative participant as that in the top panel. From this participant, 38 motor units could be decomposed and tracked across 46 valid no-go trials. The composite spike train (summing the activity of all identified motor units) was obtained and then a continuous wavelet transform analysis was used to analyse the time–frequency changes in the composite spike train

signal. The results reflect important changes in power at 3 frequencies (around 8 Hz, 20 Hz and 46 Hz). In the average results across participants, significant changes could only be found at frequencies above 15 Hz (ref. 65). Colour scales were adapted to 3 frequency ranges (5–15 Hz, 15–30 Hz and 30–60 Hz). These results motivate the investigation of high-frequency signals in MNs as a potential resource for indirectly studying changes in the CNS (data from ref. 65). **b**, Results from a study⁶⁷ combining magnetoencephalography with EMG to analyse the corticomuscular transmission of beta activity during sustained contractions. The figure shows the averaged segments of brain recordings in which beta bursts are detected (top) and the corresponding activity in the muscles (second panel), which aligns with the beta modulations in the brain. The coherence measurements between brain and muscle signals (third panel) provide consistent evidence of the common origin of the beta activity measured in the brain and muscle. Finally, the bottom panel shows how forces are kept unaltered during periods of strong beta modulation at the level of the brain and muscle (the plot shows the average and s.d. of the measured forces across all trials considered to generate these plots). In all four plots in panel **b**, Δ refers to changes in the measured magnitude. a.u., arbitrary units. Panel **b** adapted from ref. 67 under a Creative Commons license CC BY 4.0.

This would allow for the separation of the behaviourally relevant and irrelevant latent factors that drive motor unit activity. Alternatively, neurofeedback on MNs may be used to train the CNS to increase the number of motor commands to muscles to reach augmented control⁸⁵.

Decoding cortical activity from wearable sensors in the limbs to monitor brain states, without direct brain recordings, may also be used to refine existing interventions that guide neural changes in the CNS through non-invasive stimulation^{86,87}. MN-based information can be used in neuromodulation interventions, such as in closed-loop brain stimulation approaches in which the information on the brain state that triggers the stimulation is obtained from a peripheral recording. This strategy may encounter problems when the delay introduced by conduction to the periphery poses major constraints on the closed-loop algorithm⁸⁸. However, in applications where this delay is not a problem, such recordings would have several advantages with respect to current closed-loop brain stimulation approaches that estimate brain states directly from brain recordings⁸⁶. First, electrical or magnetic stimulation of the CNS produces large artefacts that strongly affect brain recordings, but they have little impact on muscle recordings that should not affect the accuracy in decoding the spiking activity

of MNs⁸⁹. Second, oscillations at high frequencies can presumably be better identified from direct neural cell spike trains, as in the case of MN decoding, than from the interference EEG signals recorded from the scalp. Finally, if oscillations in a frequency band decoded from MNs are caused only by cortical projections, or if the contributions from different sources to a given frequency band of the spectral contents in MN activity can be separated (for example, by using separation methods), then it would not be necessary to separate multiple cortical sources that oscillate simultaneously at overlapping frequencies. In fact, in this context MNs are expected to provide reliable estimates of modulations in cortical rhythmic activity based on simulations of corticospinal projections to MNs.

Considerations and limitations

This Perspective underscores recent strides made in the analysis and interpretation of muscle recordings, aiming to motivate further exploration into harnessing the information encoded by MNs about CNS activity. The promise of muscle-based neural interfaces is evident in their practicality and potential for real-world applications⁷⁶. However, it is important to consider the current state of techniques and systems

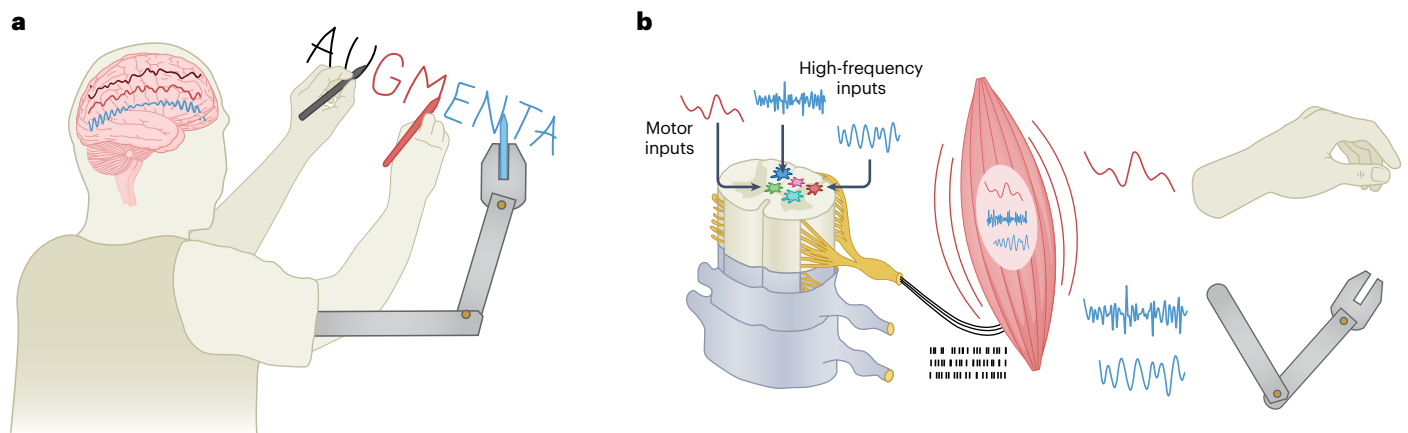


Fig. 7 | Human motor augmentation extending the number of control signals extracted from human muscles. In the field of human motor augmentation, a key question relates to the extraction of neural information from the brain that allows humans to control artificial effectors. **a**, Schematic of limb augmentation, inspired by Stelarc's third-hand art project (stelarc.org). **b**, The spinal cord projects multiple signals to the MN pools innervating muscles. Inputs common to all MNs in a muscle and outside the motor band are linearly transmitted to

muscles and are partly associated with ongoing brain activity. These high-frequency inputs to MNs and muscles may be used to augment the degrees of freedom that humans can use to control natural limbs and artificial effectors. In this view, the electric fields that can be measured over the human body reflect the activity of muscles that can be tracked back to cortical brain signals, thereby yielding a brain interface from peripheral sensors placed on limbs.

used to extract MN information from muscles, such as the reliance of MN-based interfaces on muscle activation. Information at the entry points of MNs becomes accessible only upon their recruitment and firing. A critical challenge facing EMG-based MN recording methods is decoding MNs during dynamic contractions. As muscles and limbs initiate movement, the stability of the shape of the motor unit action potential diminishes, rendering linear decomposition methods unreliable for identifying and tracking single MNs. While recent literature suggests various approaches to address this limitation^{18,22,24}, non-invasive systems capable of tracking MNs during unrestricted movements are yet to be realized, constraining the applicability of MN-based neural interfaces beyond laboratory settings. Moreover, to accurately estimate neural inputs to MNs from their outputs during dynamic conditions, it is imperative to reliably model neuromodulatory processes at the MN membrane. Such processes may transiently disrupt the linear input–output curves of MNs. Although these fluctuations may not substantially impede the extraction of common inputs from an MN pool across different frequencies, they could introduce noise, compromising the fidelity with which modulations in various common inputs are tracked^{37,90}. At present, we still do not understand the functional roles of high-frequency inputs to MNs with cortical origin, such as beta and gamma activity, nor their link with motor control. Therefore, it is difficult to predict the range of applications that may be developed by having access to this type of information from MNs. Further research is needed to explore how MNs can give a handle on the high-frequency inputs.

Conclusion

In this Perspective, we argue that the electric fields generated by muscles in our bodies may provide an indirect observation of neural activity that originates in the CNS and is transmitted through MNs and muscles without directly modulating forces. These fields reflect the behaviour of neurons that constitute the output layer of the neural network formed by the nervous system. As at least part of the information that arrives at the output of the nervous system is known to originate in regions of the CNS such as the cortex, recording from muscle tissue may allow us to establish a type of human interfacing not only with the peripheral nervous system but also with the CNS. The presence of cortical oscillations in the activity of MNs opens opportunities of interfacing with applications that range from neurofeedback to human movement augmentation.

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Author contributions

J.I. and D.F. drafted the paper. J.I., B.Z., E.B., S.N.B., C.M. and D.F. contributed to literature review, conceptualization and editing of the article.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence should be addressed to Jaime Ibáñez or Dario Farina.

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